Travelling Salesman Problem (TSP)

- Given a set of cities and distance between every pair of cities, the problem is to find the shortest possible route that visits every city exactly once and returns to the starting point.

Problem

- Let G (V, E) be a directed graph with edge costs c_{ij}
- The variable c_{ij} is defined such that $c_{ij}>0$ for all i & j and $c_{ij}=\infty$ if $<i,j> \notin E$
- Let |V| =n and assume n>1
- A tour of G is a directed simple cycle that includes every vertex in V.
- The cost of a tour is the sum of the cost of the edges on the tour.
- The travelling salesperson problem is to find a tour of minimum cost.
- **Solution**
- A tour- A simple path that starts and ends at vertex 1.
- Every tour consists of an edge <1,k> for some k $\in$ V - {1} and a path from vertex k to 1
- The path from vertex k to vertex 1 goes through each vertex in V - {1, k} exactly once.
- If the tour is optimal, then the path from k to 1 must be shortest k to 1 path going through all vertices in V - {1, k}, Hence the principle of optimality holds.
- Let **g (i, S)** be the length of a shortest path starting at vertex i going through all vertices in S, and terminating at vertex 1.
- The function g (1, V - {1}) is the length of an optimal salesman tour.

- From the principle of optimality it follows that

$$g(1, V-\{1\}) = \min_{2 \le k \le n} \{c_{1k} + g(k, V - \{1, k\})\} \qquad (1)$$

Generalizing (1) we get (for i € S)

$$g(i, S) = \min_{j \in S} \{ c_{ij} + g(j, S - \{ j \})\} \qquad (2)$$

- **g** (1, V- {1}) can be solved if we know g (k, V- {1, k}) for all choices of k.
- The g values can be obtained by using (2)

$$g(i, \phi) = c_{i1}, \ 1 \le i \le n$$